

Supplement to Radix-Packed Multi-Record Retrieval over an XPIR-Style RLWE Interface

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This supplement preserves experiment-specific summaries and formal reliability distinctions. Machine-readable values, provenance and reproduction commands are supplied separately in the artifact.

1 Native Feasibility Profiles

Table 1: Tested native profiles. All nine satisfy radix capacity at $w = \alpha\rho_0$ and the zero-noise arithmetic screen. Encrypted checks were executed only for supported profiles.

α	ρ_0	w	Largest weight	Native API	Passing cases
2	8	16	2^8	Supported	82
2	12	24	2^{12}	Supported	82
2	16	32	2^{16}	Supported	82
3	8	24	2^{16}	Supported	10
3	12	36	2^{24}	Supported	10
3	16	48	2^{32}	Weight limit	Not executed
4	8	32	2^{24}	Supported	10
4	12	48	2^{36}	Weight limit	Not executed
4	16	64	2^{48}	Weight and width	Not executed

Each two-record profile has 82 cases; each supported additional multiplicity has ten. These are finite functional checks, not latency samples or a failure-probability certificate.

2 Independent Repeated-Task Reliability

Composition of task failure probabilities. For two independent fresh-key single-record executions with actual ideal failure probabilities d_1, d_2 , the two-record task has failure

$$d_{\text{rep}} = 1 - (1 - d_1)(1 - d_2) \leq d_1 + d_2.$$

If only bounds $d_i \leq \varepsilon_i \leq 1$ are known, then $d_{\text{rep}} \leq 1 - (1 - \varepsilon_1)(1 - \varepsilon_2) \leq \varepsilon_1 + \varepsilon_2$; equality with the bounds is not asserted. For identical actual rates this is $1 - (1 - d_1)^2$. The two executions share one complete-task reliability budget.

3 Complete Cell Summaries

Primary and multiplicity intervals are pointwise 95% paired-bootstrap intervals. All experiments remain separate. Long-table values are rounded to six decimals; other tables use their displayed precision, and the machine-readable archive retains full precision.

3.1 Primary Two-Record Comparison

Table 2: All primary cells.

N	ℓ	ρ_0	View	Pairs	Median R	Interval
256	256	8	cold	30	1.099144	[1.066445, 1.139836]
256	256	8	online	30	1.068951	[1.048073, 1.117916]
256	512	8	cold	30	1.080886	[1.061159, 1.102893]
256	512	8	online	30	1.079077	[1.061351, 1.130738]
256	1024	8	cold	30	1.082284	[1.042836, 1.102549]
256	1024	8	online	30	1.100839	[1.077044, 1.118001]
256	2048	8	cold	30	1.086301	[1.067735, 1.121114]
256	2048	8	online	30	1.093787	[1.070273, 1.123688]
1024	256	8	cold	30	1.124530	[1.093311, 1.163800]
1024	256	8	online	30	1.130668	[1.116362, 1.169740]
1024	512	8	cold	30	1.139262	[1.085824, 1.191560]
1024	512	8	online	30	1.150107	[1.135753, 1.178071]
1024	1024	8	cold	30	1.158766	[1.087148, 1.181521]
1024	1024	8	online	30	1.171442	[1.131327, 1.222639]
1024	2048	8	cold	30	1.168316	[1.106926, 1.219878]
1024	2048	8	online	30	1.173708	[1.143952, 1.206580]
4096	256	8	cold	20	1.184618	[1.143132, 1.207677]
4096	256	8	online	20	1.189773	[1.132678, 1.247635]
4096	512	8	cold	20	1.151662	[1.115323, 1.222798]
4096	512	8	online	20	1.152929	[1.117815, 1.178396]
4096	1024	8	cold	20	1.176705	[1.152677, 1.207240]
4096	1024	8	online	20	1.129789	[1.089693, 1.185140]
4096	2048	8	cold	20	1.156052	[1.119723, 1.201635]
4096	2048	8	online	20	1.170338	[1.124527, 1.198264]
16384	256	8	cold	10	1.097684	[1.051781, 1.144316]
16384	256	8	online	10	1.113259	[1.085357, 1.155874]
16384	512	8	cold	10	1.123657	[1.083934, 1.186945]
16384	512	8	online	10	1.116719	[1.060667, 1.190064]
16384	1024	8	cold	10	1.120168	[1.108435, 1.202300]
16384	1024	8	online	10	1.092609	[1.047320, 1.158751]
16384	2048	8	cold	10	1.164486	[1.137415, 1.195119]
16384	2048	8	online	10	1.145828	[1.095961, 1.151669]
256	256	12	cold	30	1.088975	[1.058036, 1.136849]
256	256	12	online	30	1.108206	[1.066795, 1.132453]
256	512	12	cold	30	1.074215	[1.068415, 1.130221]
256	512	12	online	30	1.075190	[1.053641, 1.120763]
256	1024	12	cold	30	1.068229	[1.048548, 1.087032]
256	1024	12	online	30	1.104433	[1.094927, 1.134692]
256	2048	12	cold	30	1.081971	[1.063132, 1.136837]
256	2048	12	online	30	1.108511	[1.083409, 1.162922]
1024	256	12	cold	30	1.185692	[1.130365, 1.244933]

N	ℓ	ρ_0	View	Pairs	Median R	Interval
1024	256	12	online	30	1.138226	[1.100399, 1.162181]
1024	512	12	cold	30	1.132697	[1.084776, 1.174627]
1024	512	12	online	30	1.153913	[1.131944, 1.191463]
1024	1024	12	cold	30	1.139853	[1.110566, 1.198914]
1024	1024	12	online	30	1.161933	[1.115377, 1.197449]
1024	2048	12	cold	30	1.106430	[1.084795, 1.180182]
1024	2048	12	online	30	1.157400	[1.129020, 1.193766]
4096	256	12	cold	20	1.143322	[1.118374, 1.198527]
4096	256	12	online	20	1.149377	[1.085574, 1.182261]
4096	512	12	cold	20	1.157269	[1.116455, 1.235647]
4096	512	12	online	20	1.161336	[1.131151, 1.197153]
4096	1024	12	cold	20	1.169182	[1.150663, 1.210176]
4096	1024	12	online	20	1.159550	[1.120526, 1.210196]
4096	2048	12	cold	20	1.183250	[1.142277, 1.192142]
4096	2048	12	online	20	1.159451	[1.142173, 1.192097]
16384	256	12	cold	10	1.132234	[1.077790, 1.185490]
16384	256	12	online	10	1.108856	[1.067430, 1.155682]
16384	512	12	cold	10	1.132980	[1.111675, 1.172235]
16384	512	12	online	10	1.115768	[1.055107, 1.209364]
16384	1024	12	cold	10	1.140379	[1.067145, 1.188172]
16384	1024	12	online	10	1.137229	[1.092244, 1.220584]
16384	2048	12	cold	10	1.155693	[1.100465, 1.236547]
16384	2048	12	online	10	1.120187	[1.050380, 1.219731]
	256	16	cold	30	1.059097	[1.035995, 1.086378]
	256	16	online	30	1.069464	[1.048631, 1.093117]
	512	16	cold	30	1.066853	[1.045440, 1.099442]
	512	16	online	30	1.090027	[1.068877, 1.117328]
	1024	16	cold	30	1.076899	[1.057050, 1.093391]
	1024	16	online	30	1.077741	[1.041416, 1.109430]
	2048	16	cold	30	1.090995	[1.062540, 1.127972]
	2048	16	online	30	1.085561	[1.054980, 1.121594]
1024	256	16	cold	30	1.134876	[1.105762, 1.157059]
1024	256	16	online	30	1.137548	[1.120192, 1.194745]
1024	512	16	cold	30	1.135593	[1.104392, 1.193561]
1024	512	16	online	30	1.164129	[1.110685, 1.218475]
1024	1024	16	cold	30	1.161892	[1.115493, 1.226226]
1024	1024	16	online	30	1.156257	[1.134098, 1.213873]
1024	2048	16	cold	30	1.163989	[1.135025, 1.208196]
1024	2048	16	online	30	1.145458	[1.104724, 1.168897]
4096	256	16	cold	20	1.154856	[1.105471, 1.184269]
4096	256	16	online	20	1.140803	[1.104840, 1.191595]
4096	512	16	cold	20	1.142025	[1.122531, 1.173517]
4096	512	16	online	20	1.140008	[1.100066, 1.155140]
4096	1024	16	cold	20	1.168571	[1.109809, 1.196394]
4096	1024	16	online	20	1.146711	[1.100757, 1.183143]
4096	2048	16	cold	20	1.149438	[1.108011, 1.181065]
4096	2048	16	online	20	1.158237	[1.122007, 1.182179]
16384	256	16	cold	10	1.110632	[1.055060, 1.193108]
16384	256	16	online	10	1.103159	[1.068378, 1.126243]
16384	512	16	cold	10	1.094205	[1.077321, 1.140992]
16384	512	16	online	10	1.129429	[1.089600, 1.146960]
16384	1024	16	cold	10	1.148467	[1.109590, 1.186938]

N	ℓ	ρ_0	View	Pairs	Median R	Interval
16384	1024	16	online	10	1.124972	[1.059891, 1.204451]
16384	2048	16	cold	10	1.132562	[1.096479, 1.160726]
16384	2048	16	online	10	1.092641	[1.069428, 1.164926]

3.2 Multiplicity Comparison

Table 3: Multiplicity experiment at $\rho_0 = 8$, $w = 8\alpha$ and $L = 1$. Each cell contains ten measured pairs. Entries are median paired ratios with pointwise 95% intervals.

N	ℓ (bits)	View	$\alpha = 2$	$\alpha = 3$	$\alpha = 4$
1024	512	cold	1.050 [1.005, 1.078]	1.098 [1.038, 1.129]	1.140 [1.119, 1.184]
1024	512	online	1.055 [1.030, 1.082]	1.092 [1.077, 1.107]	1.151 [1.106, 1.235]
1024	2048	cold	1.036 [1.006, 1.056]	1.114 [1.078, 1.137]	1.154 [1.117, 1.186]
1024	2048	online	1.033 [0.980, 1.067]	1.096 [1.074, 1.130]	1.121 [1.088, 1.143]
4096	512	cold	1.025 [1.000, 1.052]	1.093 [1.039, 1.106]	1.122 [1.095, 1.144]
4096	512	online	1.024 [0.998, 1.039]	1.065 [1.043, 1.083]	1.108 [1.094, 1.138]
4096	2048	cold	1.013 [0.986, 1.026]	1.071 [1.061, 1.080]	1.129 [1.110, 1.152]
4096	2048	online	1.015 [0.993, 1.034]	1.066 [1.053, 1.078]	1.117 [1.097, 1.158]

The two-record entries are internal references for this experiment. Packed execution is serial; repeated tasks use α concurrent workers. This is not a fixed-total-core comparison.

3.3 Two-Record Overlap Across Experiments

The following matched workload shapes have distinct experimental contracts. Their samples are not pooled, and the difference has no established causal attribution.

N	ℓ	View	Primary R [interval]	Multiplicity R [interval]
1024	512	cold	1.139262 [1.085824, 1.191560]	1.050424 [1.005370, 1.078139]
1024	512	online	1.150107 [1.135753, 1.178071]	1.055142 [1.029526, 1.082148]
1024	2048	cold	1.168316 [1.106926, 1.219878]	1.035751 [1.005946, 1.056287]
1024	2048	online	1.173708 [1.143952, 1.206580]	1.032994 [0.980469, 1.066962]
4096	512	cold	1.151662 [1.115323, 1.222798]	1.024721 [0.999873, 1.052480]
4096	512	online	1.152929 [1.117815, 1.178396]	1.024121 [0.998385, 1.038569]
4096	2048	cold	1.156052 [1.119723, 1.201635]	1.013180 [0.985558, 1.026105]
4096	2048	online	1.170338 [1.124527, 1.198264]	1.014741 [0.993317, 1.033899]

3.4 Sensitivity and Capacity

Table 5: Bundle sensitivity at $N = 1024$, $\ell = 512$, $\alpha = 2$, $\rho_0 = 8$. Median paired ratios and ranges of three session medians; 30 measured pairs per cell.

Extraction bundle	Resource bundle	Cold	Online
Record-field	Two-CPU	1.217 [1.190, 1.246]	1.218 [1.207, 1.218]
Record-field	Four-CPU parent	1.160 [1.138, 1.193]	1.170 [1.101, 1.219]
Full-block	Two-CPU	1.209 [1.204, 1.274]	1.236 [1.208, 1.268]
Full-block	Four-CPU parent	1.168 [1.120, 1.185]	1.123 [1.091, 1.237]

Resource labels describe allowances, not packed arithmetic parallelism. Brackets are observed ranges, not confidence intervals. Implementations also differ in API wrappers; resource bundles include affinity and memory limits.

Table 6: Capacity experiment: $N = 1024$, $\alpha = 4$, $\rho_0 = 8$, $w = 32$, full-block extraction and the four-CPU parent allowance. Entries are median paired ratios with observed ranges of three session medians.

ℓ (bits)	J	L	u		Cold	Online
4096	512	1	0.5	1.339	[1.312, 1.376]	1.308 [1.232, 1.360]
8192	1024	1	1	1.311	[1.284, 1.318]	1.271 [1.268, 1.286]
12288	1536	1	1.5	1.311	[1.277, 1.336]	1.287 [1.287, 1.299]
16384	2048	1	2	1.319	[1.293, 1.337]	1.258 [1.257, 1.266]
65536	8192	2	8	1.294	[1.275, 1.310]	1.305 [1.293, 1.310]

Each cell has 30 measured pairs. Brackets are descriptive session ranges, not confidence intervals. Query buffers are 128/512 MiB for packed/repeated; reply buffers are 128/512 KiB for $L = 1$ and 256/1024 KiB for $L = 2$.